Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18EEDE541]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

ELECTRICAL MACHINE DESIGN

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Mention any six factors which impose limitations on the design of electrical machines. How do they affect design? Write briefly. **06**

b. Discuss the properties of electrical insulating materials and also provide the classification of insulating materials for the temperature range 90°C to 155°C with two examples for each. **08**

2. a. Prove that Volts/turn of a single phase transformer is given by $K\sqrt{Q}$ where Q is the output of the transformer. **06**

b. Determine the main dimensions of core, number of turns, cross sectional area of conductors of a 125 kVA, 6600/460 V, 50 Hz, 1 phase core type distribution transformer. The maximum flux density in the core is 1.2 Tesla, current density is 250 A/cm^2. Assume a cruciform core with 8% insulation between laminations. Take yoke cross section as 15% greater than that of core. The height of window is 3 times of the width of window. Net cross section of copper in window is 0.23 times the net cross sectional area of the iron in the core. Take K_w = 0.3. **08**

3. a. A design is required for a 50 kW, 4 Pole, 600 rpm, DC shunt generator. The full load terminal voltage is 220 V. If the maximum gap density is 0.83 wb/m^2 and the ampere conductors per meter is 30000. Calculate the suitable dimensions of armature core to give square pole face. Assume that the full load armature voltage drop is 3% of rated terminal voltage and that of field current is 1% of rated full load current. Ratio of Pole arc to Pole pitch is 0.67. **08**

b. Explain Specific loadings and the usual range of values for DC machines. **06**

OR

4. a. Draw the developed armature winding diagram of a DC machine with 6 poles and 12 slots. Double layer, the winding is to be lap wound. Mark the poles and show the brush positions. **10**

b. Define the following terms with respect to DC winding. **04**

- **i.** Back pitch
- **ii.** Front pitch
- **iii.** Winding pitch
- **iv.** Commutator pitch

5. a. A small induction motor has one conductor per slot. If there are 24 slots, and the stator is wound for 4 poles, draw the winding diagram, connected as wave winding. Make the end connection for star operation. **14**

OR

6. a. Compute the main dimensions, number of stator slots, number of stator conductors and the number of turns per phase of a 3.7 kW, 400 V, 3 phase, 4 pole, 50 Hz squirrel cage induction motor to be started by a star delta starter. Assume average flux density in air gap = 0.45 wb/m^2, ampere conductors per meter = 23000, efficiency = 0.85, power factor = 0.84 and therefore choose the main dimensions to obtain the good overall design. **10**

b. Draw the speed torque characteristics of an induction motor and explain the phenomena which are observed due to wrong combination of number of rotor and stator slots. **04**

7. a. Along with nomenclature, derive the output equation of synchronous machine **07**

b. Define SCR of a synchronous machine. How does it affect the performance of machine? **07**

OR

8. a. For a 250 kVA, 1100 V, 12 pole, 500 rpm, 3 phase alternator, determine the air gap diameter (stator conductors, number of stationary slots and cross section of stator conductors. Average gap density = 0.6 T. Specific electric loading 30000 ampere conductors/meter, $L/\tau = 1.5$, $\delta = 3.5$ A/mm^2, $k_w = 0.955$. **14**